Full Title of the Thesis, One or two lines

Student name

[Pre-Defence / Final Defence]

[MS / PhD] [Name of degree]

Supervisor: [Prof.] Dr. [Name of supervisor]

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National University of Sciences and Technology (NUST), Islamabad, Pakistan

September 21, 2026

Outline

1. Introduction
2. Literature Review
   - Research Gaps Identified
3. Problem Statement
4. Research Questions
5. Research Objectives
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   - Alternative Approaches
   - Details
7. Implementation
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   - How Research Questions are Answered and Research Objectives are Met
9. Conclusion
10. Misc Link (Git/Publication)

Background: [from the established picture to the proposed one]

[The established approach]

- what it does, in one line; an equation if there is one: $x_{t+1} = f(x_t)$
- its characteristic behaviour
- where it is used

[The proposed approach]

- what changes, with a citation [1]
- the defining relation,

$$U = S\,(\mathbb{I} \otimes C)$$

- its characteristic behaviour, contrasted with the left column

Why this is a candidate [general tool / better method]

One or two sentences, each supported by a citation [2, 3], giving the reason the proposed approach deserves the study.

Motivation

- **First motivating fact.** State it and say why it matters.
- **Second motivating fact.** The step from the first to the problem studied.
- **Third.** What changes when the proposed approach is adopted, with a citation [1].

The practical interest

Where the results would be used in practice, and why a method with these properties is plausibly a common ingredient of those uses.

Literature review (1): [first parent field]; [second parent field]

[First field]

- key result one [1]
- key result two [3]
- key result three
- foundational theorems [2]

[Second field]

- key result one [1]
- key result two; the known obstacle [3]
- benchmarking and the state of the evidence [2]

Literature review (2): [the intersection of the two fields]

A considerably smaller body of work than either parent field.

- **[Closest antecedent one.]** What it did [1] — and in what respect it differs from this work.
- **[Closest antecedent two.]** What it did [3] — and what it is *not*.
- **[Design choices that matter and are seldom varied.]** Evidence that a choice usually fixed by convention changes the outcome [2].

Research gaps identified

Gap 1 — [short name]

What the literature asserts or assumes, and what it has not actually done.

Gap 2 — [short name]

Which free choices are fixed by convention, with evidence that they matter [1].

Gap 3 — [short name]

What comparison is missing, and under what protocol it would have to be made [3].

Problem statement

- The observation the problem rests on: **what is seldom varied, measured, or compared** in existing work.
- Why the obvious justification for the proposed approach is not by itself an answer.

The problem addressed

One sentence stating exactly what is instantiated, on what terms, across which cases — **[case family A]** and **[case family B]** — and what is measured for each.

Framing: the one-line framing of the thesis, and what remains the application.
Limits fixed at the outset: what is and is not claimed; the simulation or experimental regime; how a negative outcome is treated.

Research questions

For every case the same three questions are posed:

RQ1 **[Formulation.]** What is asked first, stated precisely.

RQ2 **[Cost.]** What is asked second, and in which units.

RQ3 **[Performance.]** What is asked third, and against what baseline.

Variables of the investigation (not its subject)

The parameters that are varied wherever a case makes the comparison meaningful, listed so the committee can see they are controls rather than the contribution.

Research objectives

O1 Produce a **[deliverable one]** on [cases].

O2 Exhibit **[deliverable two]**, simulate it, and cost it.

O3 Show that **[claim three]**, and evaluate it against [baselines].

O4 Reproduce **[known result]** and quantify [what it costs].

O5 Build **[the learning / application models]** and evaluate them under a fixed protocol against tuned baselines.

O6 Report negative results with the same precision as positive ones, and identify the **regime in which the approach is advantageous**.

Justification of the methodology

- **Why [the central construction] everywhere?** The reason, with a citation [1]; where the alternative construction is still used and why. (Note any committee observation from the proposal defence that this answers.)
- **Why one primitive across [several] cases?** What that is the only way to test.
- **Why [simulation / this experimental setting]?** What the setting guarantees and what it leaves out.
- **Why [this reporting discipline] [3]?** Tuned baselines, splits fixed before model selection, results over seeds.
- **Why report cost in [these units]?** What the more common unit hides.

Alternative approaches considered

Alternative	Why it was not the main route
[Alternative one] [1]	Why it does not answer the research questions; what is built instead.
[Alternative two]	Its cost or prerequisite; where it is nevertheless used.
[Standard constructions from the literature]	Kept — but as tuned *baselines*, not as the object of study.
[Runs on real hardware / field data]	Not needed to answer RQ1–RQ3; what would obscure the comparison. Future work.
[Out-of-scope generalisation] [2]	Out of scope; the restriction that applies throughout.

Details (1): [the central object as a design space]

Setting. The objects, their sizes, and the notation used for the rest of the talk.

The family used throughout:

$$K(\delta, \theta) = \Phi(\delta)\, e^{-i\theta G}$$

- what the first parameter interpolates between
- what the second parameter interpolates between
- a property that makes it usable in the method (e.g. differentiable)

Two axes of variation

1. **[Axis one]**: what it depends on; its cost.
2. **[Axis two]**: what it depends on; parameter count; the axis the trained models use.

Characterised before any objective is optimised: the quantities measured for every configuration.

Details (2): [first group of cases] — construction, baseline, protocol

Case	Construction	Baseline	Protocol
[Case A]	what is built	what it is compared with	sizes, seeds, tooling, repetitions
[Case B]	what is built	what it is compared with	sizes, seeds, tooling, repetitions
[Case C]	what is built	what it is compared with	sizes, seeds, tooling, repetitions
[Case D]	what is built	what it is compared with	sizes, seeds, tooling, repetitions

Details (3): [second group of cases] — construction, baselines, protocol

Case	Construction	Baselines	Protocol
[Case E]	what is built	the baselines, tuned [1]	dataset, split, loss, optimiser, budget, seeds
[Case F]	what is built	the baselines, tuned	datasets [3]; cross-validation; splits
[Case G]	what is built	the baselines, tuned [2]	datasets; folds

One sentence on the protocol rule that applies to every row (e.g. hyperparameters selected on the training fold alone).

Implementation

Tooling

- libraries and versions
- any custom core, and how it was cross-checked
- validation against an independent reference

Reproducibility

- every stochastic element seeded and stated
- results written to a record; every table and figure generated from it by script

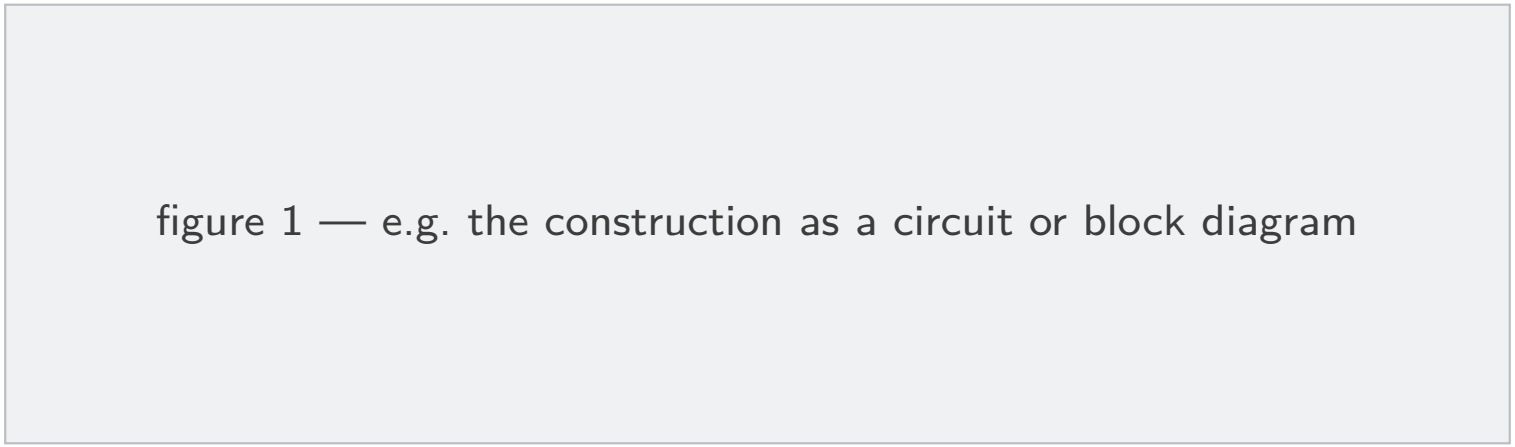
Code organisation (`[repository]`)

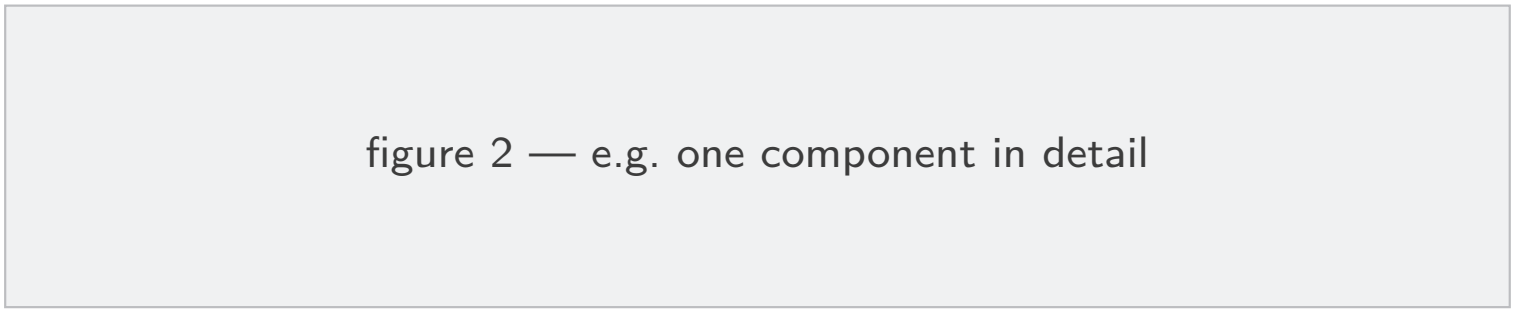
- `[module one]` — what it holds
- `[module two]` — what it holds
- `[module three]` — what it holds

Limits

- the size or regime that is tractable, and what sets it
- what is idealised away, and what that makes the reported numbers (e.g. a ceiling)

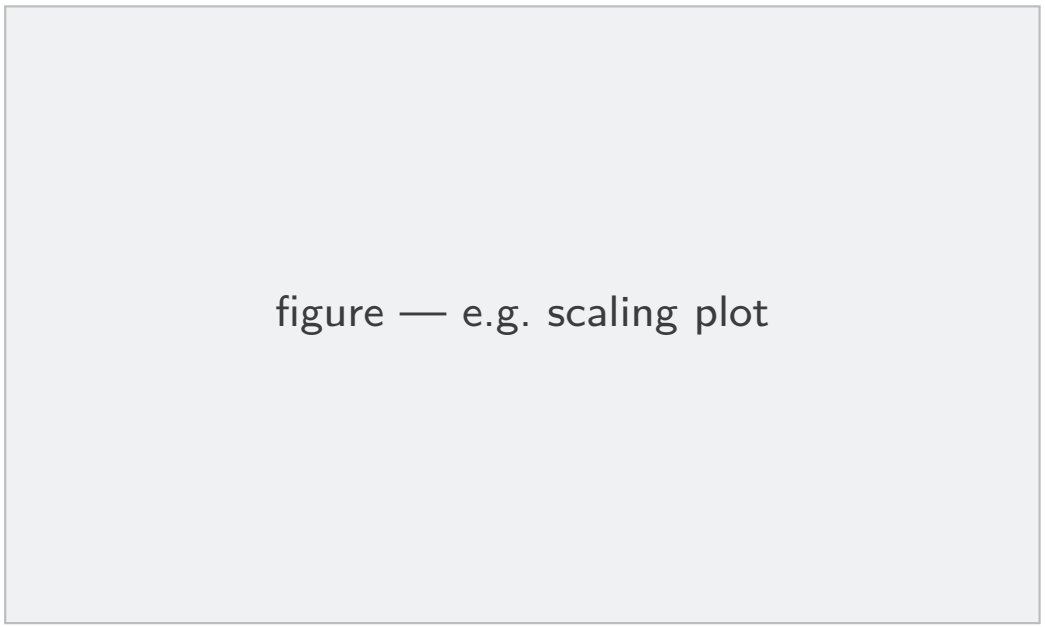
Implementation: [the central construction as a diagram]





- What one layer / unit of the construction consists of.
- The feature of the construction the results will turn on.

Result 1 — [what was measured] (RQ2)

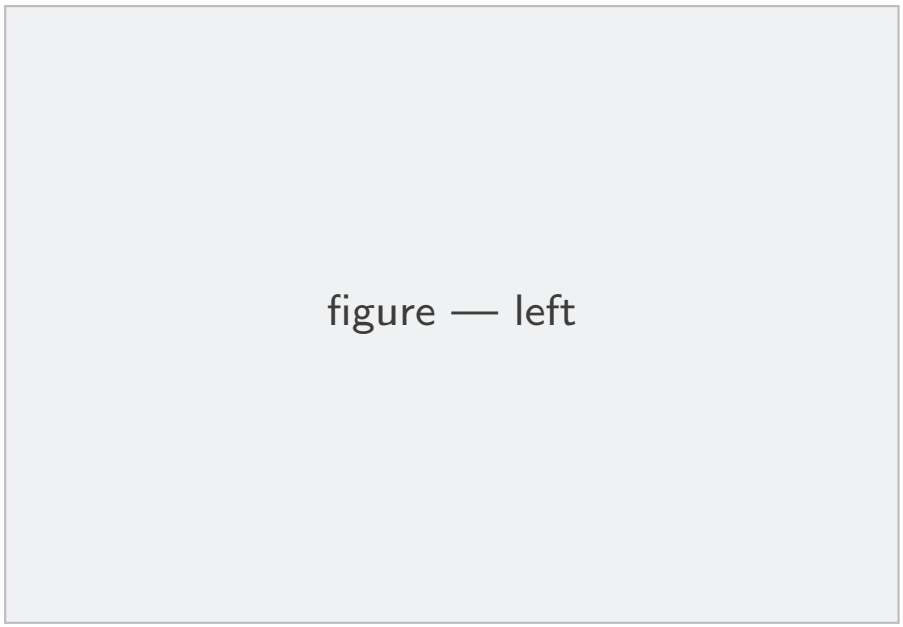


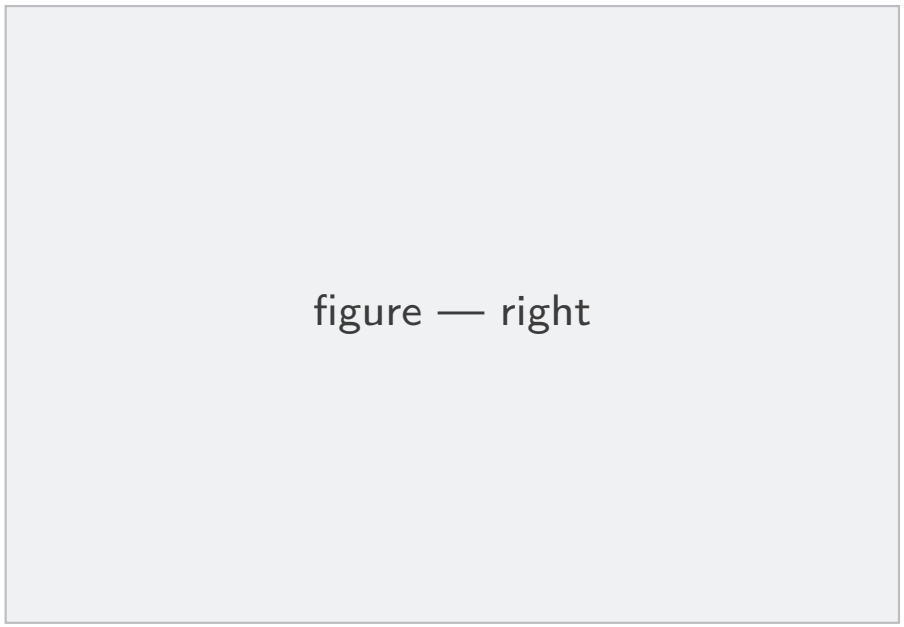
Case	N	Size	Baseline	Proposed
A1	64	7	7 270	422
A2	256	9	119 383	902
A3	1024	11	—	1 614
B1	64	8	29 655	153
B2	256	10	—	257

- baseline: its scaling
- proposed: its scaling — the headline saving
- the variant that does *not* scale, and by how much

The condition under which the saving holds, and where it does not.

Result 2 — [what was measured] (O4)





- The reproduced known result, with the fitted exponent and the theorem it agrees with [1].
- The extension, and what it costs.
- The one-sentence conclusion the committee should take from this frame.

Result 3 — [what was measured] (O2)

figure

N	a	r	$P(r)$	cost	outcome
15	7	4	0.51	522	3×5
21	2	6	0.31	10 103	7×3
35	3	12	0.32	49 739	7×5

- what the construction is, in one line
- what it achieves, quantitatively
- what it does not give, stated as plainly as the positive result

Result 4 — [what was measured] (O3)

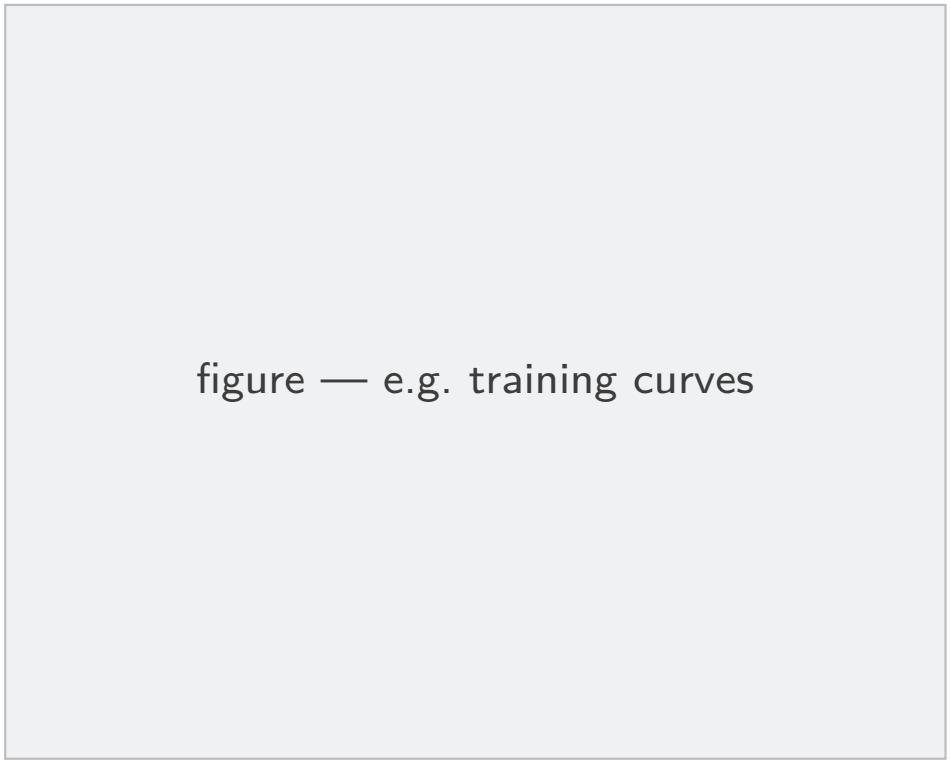
figure

Setting; figure of merit defined in one line

p	Method	Ratio	$P(\text{ok})$
1	proposed (variant 1)	0.852 ± 0.022	1.00
	proposed (variant 2)	0.759 ± 0.018	1.00
	baseline	0.308 ± 0.066	0.38
4	proposed (variant 1)	0.910 ± 0.020	1.00
	proposed (variant 2)	0.857 ± 0.051	1.00
	baseline	0.370 ± 0.077	0.47

- the construction and the identity or property it rests on
- the headline: $> 2\times$ the baseline — and *why*
- honest reading: where the proposed method is weaker

Result 5 — [first learning / application result] (O5)



Task; metric over k repetitions

Model	Size	Params	Acc. (%)
reference model A	8	16	99.0 ± 3.0
proposed (best variant)	**5**	32	$\mathbf{99.0 \pm 2.0}$
proposed (variant 2)	5	32	92.7 ± 4.4
proposed (variant 3)	5	10	92.5 ± 3.0
reference model B	8	16	71.3 ± 11.2

- the reference reproduced, and what its result depends on
- the proposed model matches it at lower size — and why
- a variant that does worse, and the interpretation
- the cost caveat that keeps this a simulation result

Result 6 — [second learning / application result] (O5)

figure — e.g. accuracy by dataset

Secondary statistic, defined in one line

Dataset	Proposed	Base. 1	Base. 2
dataset 1	0.79	0.79	0.35
dataset 2	0.81	0.89	0.28
dataset 3	0.69	0.68	0.10
dataset 4	0.60	0.60	0.10

- a sanity check that passed
- the headline: best or joint-best among the like-for-like methods
- the known failure mode [1] that the proposed method avoids, and why
- no advantage over the tuned classical baselines: say so

Result 7 — [third learning / application result] (O5)

figure

Metric (%), $k \times m$ folds

Method	Set 1	Set 2	Set 3	Set 4
proposed	84.2 ± 8.8	57.0 ± 7.0	74.6 ± 3.7	70.6 ± 3.5
classical control	87.2 ± 8.3	58.5 ± 9.9	74.8 ± 4.3	70.6 ± 4.9
standard baseline	82.0 ± 10.3	60.3 ± 6.4	75.0 ± 3.8	71.0 ± 3.9
proposed, trained	88.3 ± 7.0	57.6 ± 6.6	—	—
proposed, fixed	88.1 ± 8.2	57.5 ± 4.9	—	—

- what the proposed method reaches, against the standard baseline
- it does not beat the classical control: say where and by how much
- an identified confound, and what must be repeated

How the research questions are answered (1): [first group of cases]

Case (obj.)	RQ1 — formulation	RQ2 — cost	RQ3 — against a tuned baseline
[Case A] (O1)	the formulation in one line	its cost	+ the positive finding; − the limit
[Case B] (O2)	the formulation in one line	its cost	the quantitative outcome; − what it does not give
[Case C] (O3)	the formulation in one line	its cost	+ the positive finding; − where it is weaker
[Case D] (O4)	the formulation in one line	its cost	matches the known result; − the qualification

O6 — every negative outcome is reported at the same precision as the positive ones.

How the research questions are answered (2): [second group of cases]

Case (obj.)	RQ1 — formulation	RQ2 — cost	RQ3 — against a tuned baseline
[Case E] (O5)	the formulation in one line	size vs. the reference; the depth caveat	+ matched the reference; – a variant fits worse; – not a device proposal
[Case F] (O5)	the formulation in one line	its cost	+ best or joint-best like-for-like; – within noise of the classical baselines
[Case G] (O5)	the formulation in one line	its cost	+ matches the standard baseline; – never beats the classical control

Regime identified (O6): the one-sentence statement of where the approach pays and where it does not.

Discussion: what [the approach] buys, and what it does not

[The approach] buys

1. **[Gain one]** — the number that shows it.
2. **[Gain two]** — the number that shows it.
3. **[Gain three]** — the number that shows it.
4. **[Gain four]**.

[The approach] does not buy

1. [Non-gain one], stated plainly.
2. [Non-gain two], with the size of the effect against the noise.
3. [Non-gain three].
4. [Non-gain four].

Pattern: the one-sentence rule that separates the two columns.

Threats to validity

- **Scale.** The sizes reached, and what the fitted quantities do and do not establish.
- **Cost proxies.** What the cost measure ignores; which comparisons are therefore claimed.
- **Optimisation.** The optimiser and budget; what was *not* measured — name the largest single gap in the evidence.
- **Data.** Provenance of the datasets; which baselines reproduce published figures; what the synthetic tasks are and are not evidence for.
- **Scope.** The restrictions per case, in one line each.

Conclusion

- What was treated as **the object of study**, and across how many cases, on what terms.
- **The central lesson**, in one sentence.
- **Where it pays:** the list, in one line.
- **Where it does not:** the list, in one line.
- The scope of the claims; **what is not claimed**.

Limitations and future work

Limitations

A single paragraph listing the restrictions of the study, separated by semicolons, with the most consequential one marked in italics.

Future work (in order of how well-posed it is)

1. The best-defined next step, with what it would cost.
2. The measurement that was missing, with the references that define it [1, 3].
3. The natural extension of the strongest result.
4. The open question that decides whether the programme continues.

Code, manuscript and publications

- **Simulation code** (what it contains):
  `[repository URL]`
- **Manuscript source** (LaTeX, bibliography, generated tables):
  `[repository URL]`
- **Conference poster / talk** — [venue, year]:
  `[title / link]`
- **Publication / preprint:**
  `[none yet / arXiv link]`

One sentence on how the figures and tables in the talk were produced and how the random components were fixed.

References I

[1] F. Author and S. Author, "Title of an example journal article," *Journal Name*, vol. 1, no. 1, pp. 1–10, 2020.

[2] S. Author, *Title of an Example Book*. City: Publisher Name, 2010.

[3] T. Author, F. Author, and F. Author, "Title of an example conference paper," in *Proceedings of an Example Conference*, 2021, pp. 100–110.

Thank you

Questions